

Interrupts & Timers

TinkerCAD Activity: "Interrupts & Timers"

* In the following virtual exercises we will not take into account the "debouncing" method discussed in the previous lessons.

Reminder: EICRA Register is used to configure the trigger condition for external interrupts.

- **EICRA (External Interrupt Control Register A):** Configures the trigger condition for external interrupts

- For each interrupt (INT0 and INT1), there are two corresponding bits (ISCn1 and ISCn0) that define the trigger condition.
- * "n" is the interrupt number
- The combination of these bits determines whether the interrupt is triggered on:
 - low level
 - logical change
 - falling edge
 - rising edge

For INT0 (bits ISC01 and ISC00):

- 00: Low level triggers the interrupt (default).
- 01: Logical change triggers the interrupt.
- 10: Falling edge triggers the interrupt.
- 11: Rising edge triggers the interrupt.

For INT1 (bits ISC11 and ISC10):

- 00: Low level triggers the interrupt (default).
- 01: Logical change triggers the interrupt.
- 10: Falling edge triggers the interrupt.
- 11: Rising edge triggers the interrupt.

12.2 Register Description

12.2.1 EICRA – External Interrupt Control Register A

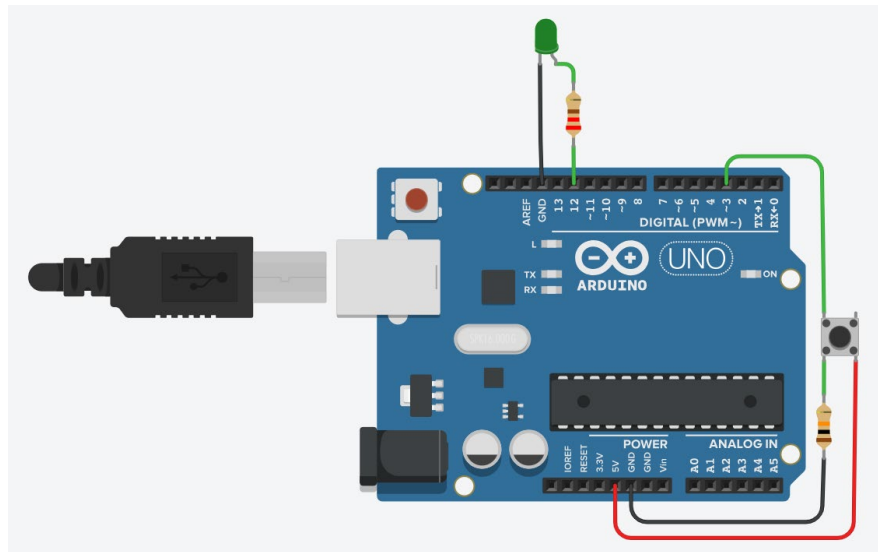
The external interrupt control register A contains control bits for interrupt sense control.

Bit (Out99)	7	6	5	4	3	2	1	0	
	–	–	–	–	ISC11	ISC10	ISC01	ISC00	EICRA
Read/Write	R	R	R	R	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Handwritten notes: Arrows point from ISC11 and ISC10 to "INT1" and from ISC01 and ISC00 to "INT0".

1. "1. Introduction to Interrupts - Toggle LED"

- a. Go to "Activities", "Copy and Tinker" the associated task, and update the file name.
- b. Construct the following structure:
 - i. Arduino Uno R3
 - ii. 1 Pushbutton on 3.
 - iii. 1 10K resistor.
 - iv. A green LED on 12.

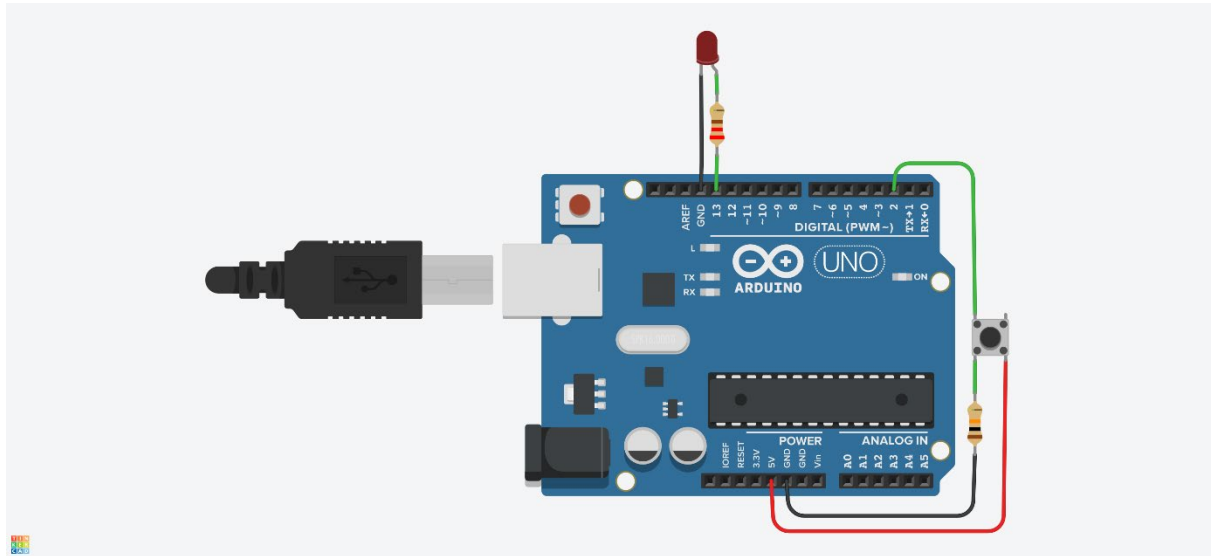


- c. Write a code that does the following:
 - i. Setting relevant pins as INPUT/OUTPUT.
 - ii. Configuring relevant interrupts related registers (based on the requirements in the code).
 - iii. The program should **toggle** the LED when the switch button is **PRESS**.

2. “2. Toggle LED on every Press/Release”

- a. Go to “Activities”, “Copy and Tinker” the associated task, and update the file name.
- b. Construct the following structure:

Make sure the pins are connected exactly as you can see on the following image:



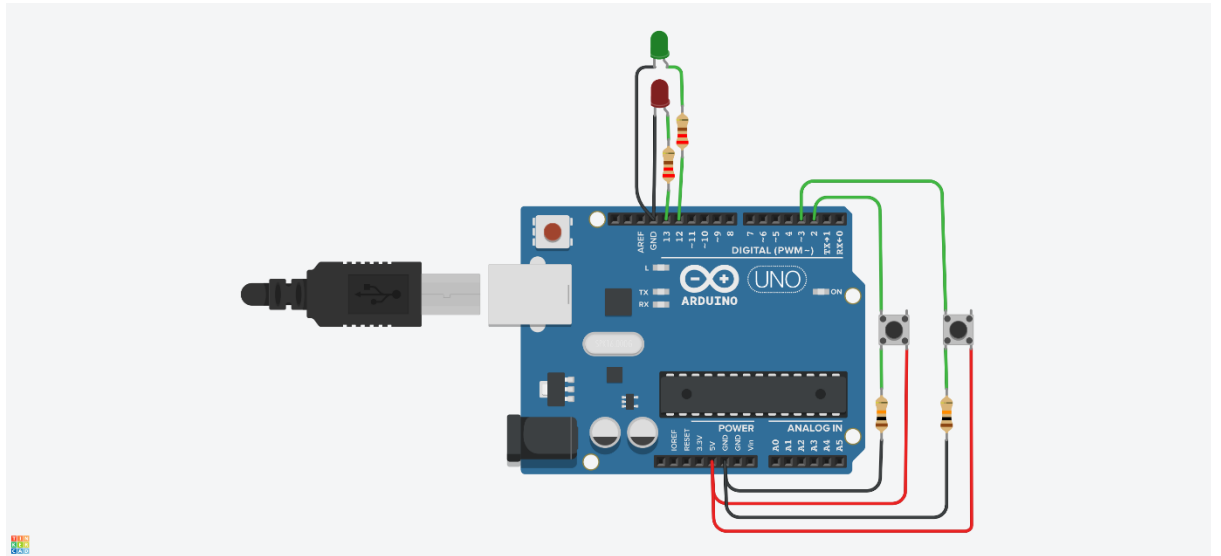
- c. Write a code that does the following:
 - i. Setting relevant pins as INPUT/OUTPUT.
 - ii. The program should **set** the **RED LED** on every **PRESS** and **clear** on every **RELEASE** (using External Interrupts).

1. PRESS: Set.
2. RELEASE: Clear.

3. "3. Configure both SW to Toggle"

- a. Go to "Activities", "Copy and Tinker" the associated task, and update the file name.
- b. Construct the following structure:

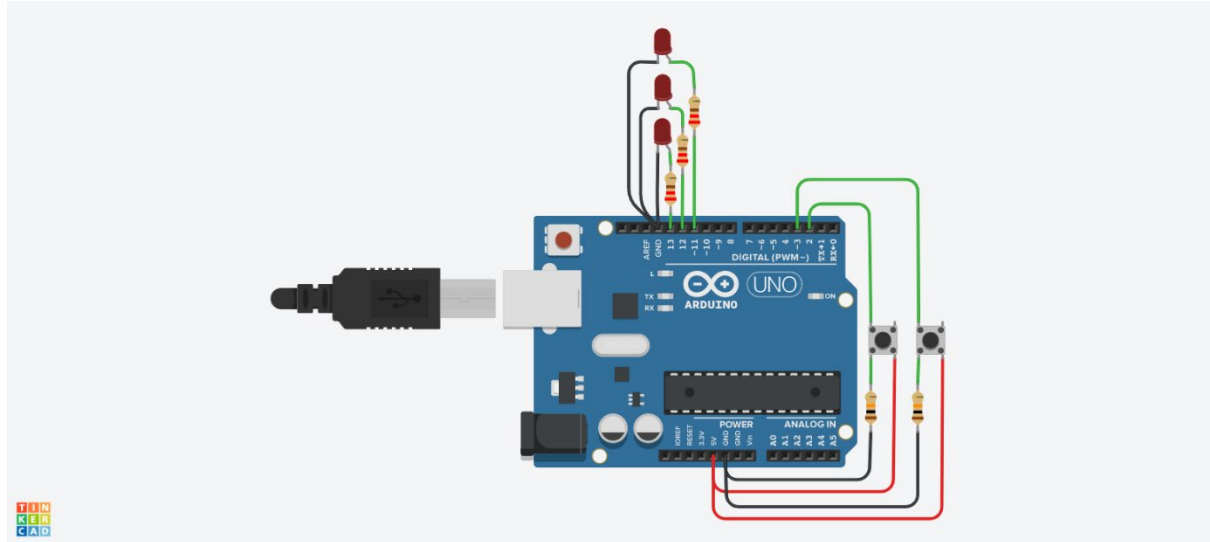
Make sure the pins are connected exactly as you can see on the following image:



- c. Write a code that does the following:
 - i. Configure Inputs/Outputs
 - ii. Configure both switches to work with interrupts:
 1. SW1 should be configured with "Rising Edge" to toggle GREEN LED.
An interrupt received by SW1 should toggle GREEN LED.
 2. SW2 should be configured with "Rising Edge" to toggle RED LED.
An interrupt received by SW1 should toggle RED LED.

4. “State Machine (with RESET button) – using Interrupts”

- Go to “Activities”, “Copy and Tinker” the associated task, and update the file name.
- Construct the following structure:



- Write a code that simulates a “state machine” defined as the following:

	Button Press / State	LED1	LED2	LED3
0	Initial State	0	0	0
1	Press #1	0	0	1
2	Press #2	0	1	0
3	Press #3	0	1	1
4	Press #4	1	0	0
5	Press #5	1	0	1
6	Press #6	1	1	0
7	Press #7	1	1	1

Note: After 7th button press, the state machine should return to the “Initial State”.

- Include a “Resetting Functionality” for SW2. When SW2 has been pressed, the state machine should return to the “Initial State”, regardless of the current state. That means when the SW2 is pressed, all the LEDs turn off and the state machine is at its “Initial State” (before Press #1).
- MAKE SURE YOU IMPLEMENT IT USING INTERRUPTS!